Worksheet 6: Giant Planets and Small Bodies

Covers Lectures 11 and 12: giant-planet interiors, the Roche limit and rings, radioactive dating, Kirkwood gaps, cometary activity

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that…"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

Constants and data

$\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $G = 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $u = 1.661 \times 10^{-27}$ kg, $S_0 = 1361$ W m^{-2} (at 1 AU), $\ln 2 = 0.6931$

Jupiter: $M = 1.898 \times 10^{27}$ kg, $R = 6.991 \times 10^7$ m, semi-major axis $a_J = 5.203$ AU, orbital period $T_J = 11.86$ yr. Saturn: $M = 5.683 \times 10^{26}$ kg, volumetric mean radius $R_p = 58{,}232$ km, bulk density $\rho_p = 687$ kg m^{-3}

Metallic-hydrogen transition near 100 GPa. Ring and moon densities: water ice $\rho_s = 1000$ kg m^{-3}, rock $\rho_s = 3000$ kg m^{-3}. Saturn A-ring outer edge observed at ~ 137,000 km

^{26}Al: half-life $t_{1/2} = 0.717$ Myr, Solar-System initial ratio $(^{26}\mathrm{Al}/^{27}\mathrm{Al})_0 = 5.2 \times 10^{-5}$. ^{238}U: $t_{1/2} = 4.4683$ Gyr; ^{235}U: $t_{1/2} = 0.7038$ Gyr; present $^{238}\mathrm{U}/^{235}\mathrm{U} = 137.82$; CAI age $t = 4567.3$ Myr

Water ice: latent heat of sublimation $L = 2.83 \times 10^6$ J kg^{-1}, molecular mass $m = 18\,u$. Cometary nucleus: Bond albedo $A \approx 0$ (a dark, fast rotator)

The uniform-density central pressure of a self-gravitating sphere in hydrostatic equilibrium is $P_c = 3GM^2/(8\pi R^4)$; the fluid Roche limit is $d_R = 2.46\,R_p(\rho_p/\rho_s)^{1/3}$ and the rigid-body value is $d_R = 1.26\,R_p(\rho_p/\rho_s)^{1/3}$. Data as in the Lecture 11 and 12 notes.

Problem 1 Giant-planet interiors and the metallic-hydrogen transition

Jupiter and Saturn are fluid hydrogen-helium planets. Their deep interiors reach pressures high enough to squeeze hydrogen into a metallic, electrically conducting state. A crude uniform-density model already shows where and why this happens.

(a) Compute Jupiter's bulk density from its mass and radius.

(b) Using the uniform-density result $P_c = 3GM^2/(8\pi R^4)$, estimate Jupiter's central pressure. Express it in GPa and compare it with the ~ 100 GPa metallic-hydrogen transition.

(c) Repeat the estimate for Saturn. Compare the two central pressures against the metallization pressure and state what this implies for the relative size of each planet's metallic-hydrogen region.

(d) Saturn radiates more energy than it absorbs from the Sun, more than slow contraction alone can supply, while Jupiter's excess luminosity is closer to what contraction gives. Explain in two or three sentences how the smaller metallic-hydrogen region you found in part (c) is linked to the extra energy source (helium rain) that Saturn needs.

Problem 2 The Roche limit and Saturn's rings

The inner edge of the region where a moon is tidally torn apart, rather than held together by its own gravity, is the Roche limit. It sets where rings can exist and where moons can survive. Saturn is the clearest case.

(a) Compute the rigid-body Roche limit for an icy body at Saturn, $d_R = 1.26\, R_p(\rho_p/\rho_s)^{1/3}$. Express it in units of R_p.

(b) Compute the fluid Roche limit, $d_R = 2.46\, R_p(\rho_p/\rho_s)^{1/3}$. Compare it with the observed outer edge of the A ring at ~ 137,000 km.

(c) A ring particle made of rock rather than ice has $\rho_s = 3000\ \mathrm{kg\,m^{-3}}$. Recompute the fluid Roche limit and explain why the main rings, which extend out to near the fluid Roche limit of part (b), are icy rather than rocky.

(d) Small moons such as Pan and Daphnis orbit inside the A ring, well inside the fluid Roche limit of part (b) but outside the rigid-body limit of part (a). Explain in two or three sentences how both facts can be true, and what the two limits bracket.

Problem 3 Dating the Solar System with radioactive decay

Meteorites carry two kinds of radioactive clock. Short-lived isotopes such as ^{26}Al are now extinct but timed the first few million years; long-lived isotopes such as uranium give absolute ages of billions of years. You will use both.

(a) For radioactive decay $N = N_0\, e^{-\lambda t}$ with $\lambda = \ln 2/t_{1/2}$, compute the decay constant of ^{26}Al and the fraction of the initial ^{26}Al left after 2 Myr.

(b) Radiogenic heating scales with the amount of live ^{26}Al present at accretion. A planetesimal that accretes 2 Myr after the first solids (CAIs) starts with the Solar-System initial $^{26}\mathrm{Al}/^{27}\mathrm{Al}$ multiplied by the surviving fraction you found in part (a); compute this ratio. State the factor by which its heating rate is lower than a body that formed at CAI time, and what this implies for melting.

(c) The Pb-Pb clock uses two uranium chains. For a sample of age t the radiogenic ratio is

$$\frac{^{207}\mathrm{Pb}^*}{^{206}\mathrm{Pb}^*} = \frac{^{235}\mathrm{U}}{^{238}\mathrm{U}}\,\frac{e^{\lambda_{235}t} - 1}{e^{\lambda_{238}t} - 1}.$$

Using the CAI age $t = 4.5673$ Gyr and present $^{238}\mathrm{U}/^{235}\mathrm{U} = 137.82$, compute this ratio.

(d) Explain in two or three sentences why the short-lived ^{26}Al clock gives high time resolution but only relative ages, while the long-lived U-Pb clock gives absolute ages but coarser resolution over the first few million years.

Problem 4 Kirkwood gaps and mean-motion resonances

The asteroid belt has gaps at specific distances where an asteroid's orbital period is a simple fraction of Jupiter's. These mean-motion resonances pump orbital eccentricities and clear the gaps. They are the Kirkwood gaps.

(a) At an interior $j : k$ resonance the asteroid completes j orbits for every k of Jupiter's, so its period is $T = (k/j)\,T_{\mathrm{J}}$. Using Kepler's third law $a^3 = T^2$ (with a in AU and T in yr, both scaled to a_{J}), show that the resonance semi-major axis is $a = a_{\mathrm{J}}\,(k/j)^{2/3}$, and compute the 3 : 1 location.

(b) Compute the locations of the 5 : 2 and 2 : 1 resonances.

(c) Using Kepler's third law, compute the orbital period of an asteroid at the 3 : 1 location you found in part (a), and confirm it is one third of Jupiter's period.

(d) An asteroid exactly at a Kirkwood gap and one just outside it look identical in a snapshot of positions. Explain in two or three sentences what physically empties the gap, and why the gaps are visible in a plot of semi-major axis but not in an instantaneous map of the belt.

Problem 5 Cometary activity and heliocentric distance

A comet is a dirty snowball that turns on as it nears the Sun and its ices sublimate. Distance is the control. Simple energy balance sets both the temperature and the gas production, and explains why comets are active only in the inner Solar System.

(a) For a dark, fast-rotating nucleus ($A \approx 0$) the whole surface reradiates the absorbed sunlight, $S_0(1\,\mathrm{AU}/r)^2 = 4\sigma T^4$. Compute the temperature at 1 AU and give the scaling $T(r)$. Evaluate it at 3 AU.

(b) Water ice sublimates efficiently once the nucleus reaches about 150 K. Find the heliocentric distance where $T(r) = 150$ K and interpret it as the onset distance for strong activity.

(c) Close to the Sun almost all absorbed sunlight goes into latent heat, so the sublimation rate per unit area is $Z = S_0(1\,\mathrm{AU}/r)^2/(4Lm)$ molecules per square metre per second. Compute Z at 1 AU and give its scaling with r.

(d) Explain in two or three sentences why comets are active only near the Sun and dormant in the outer Solar System, and why the true fall-off in gas production beyond a few AU is much steeper than the r^{-2} of part (c).